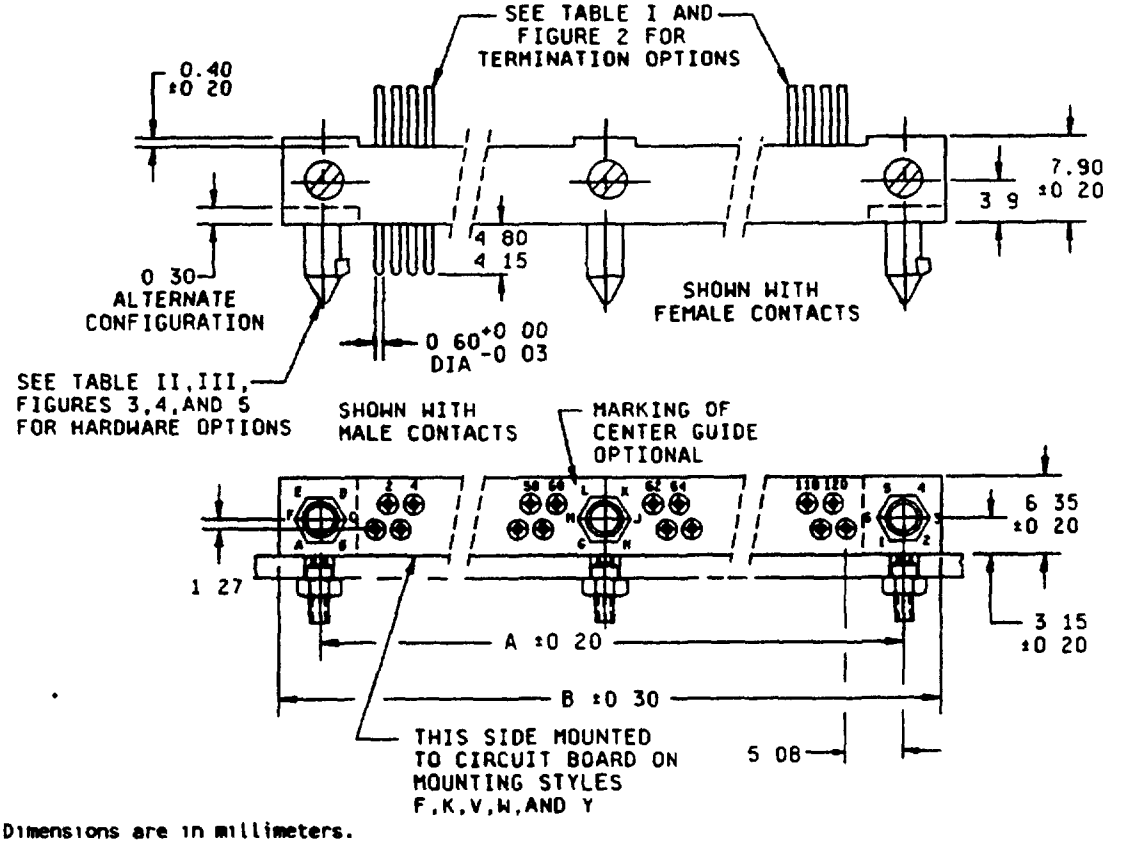
MILITARY SPECIFICATION SHEET

# CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES, PLUG, 72, 84, 96, AND 120 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS, 2.54 MM X 1.27 MM OFFSET GRID, METRIC

This specification is approved for use within the US Army Communications - Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55302.




NOTE: Dimensions are in millimeters.

FIGURE 1. Connector receptacle.

TABLE I. Number of contacts.

| Number of contacts | Dimension (in millimeters) | |
|---|---|---|
| | A | B |
| 72 | 106.70 | 114.70 |
| 84 | 121.90 | 129.90 |
| 96 | 137.20 | 145.20 |
| 120 | 167.60 | 175.60 |


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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# RECEPTACLE TERMINAL STYLES

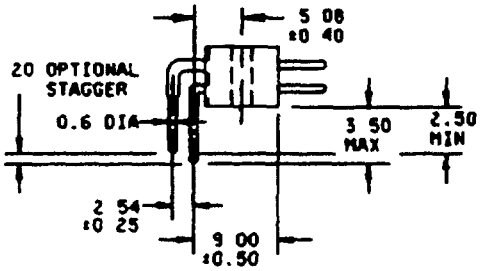
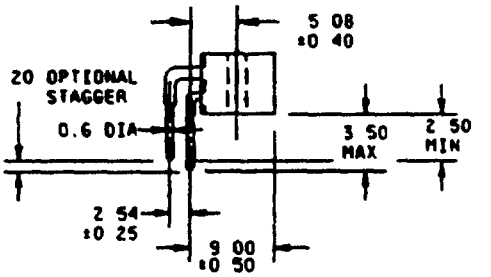
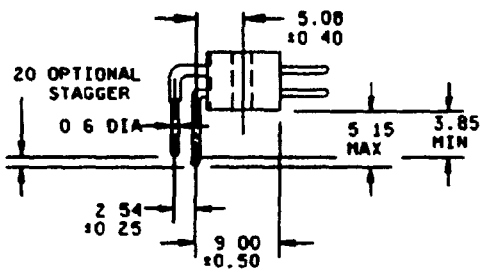
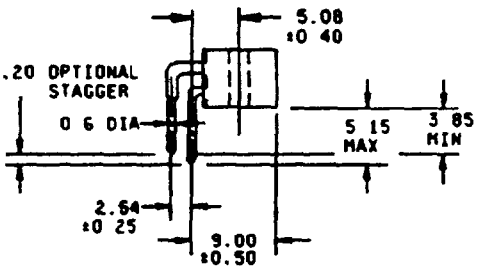
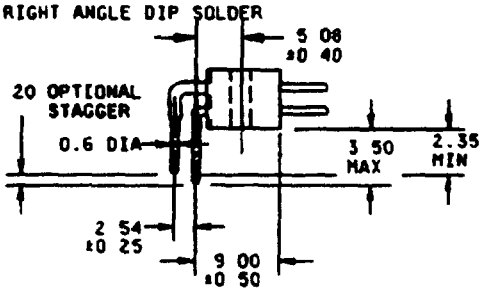
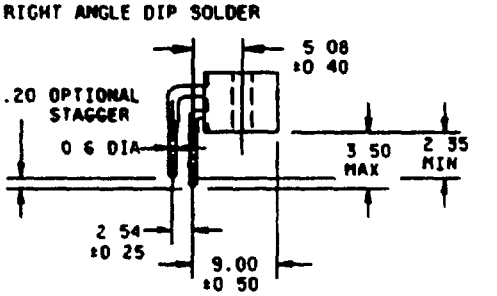
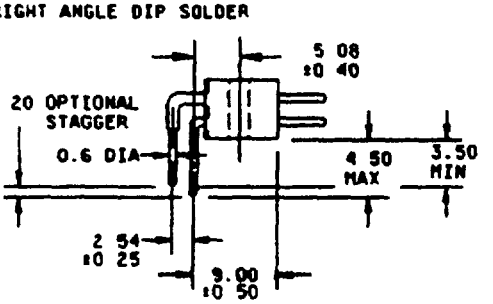
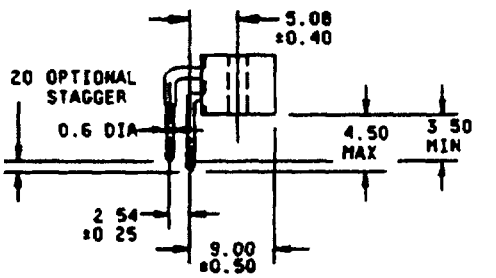
| REF | MALE CONTACTS | REF | FEMALE CONTACTS |
|---|---|---|---|
| C | RIGHT ANGLE DIP SOLDER<br>  <br><br>2 4 ( 094 IN ) P C BOARD | E | RIGHT ANGLE DIP SOLDER<br>  <br><br>2 4 ( 094 IN ) P C BOARD |
| B | RIGHT ANGLE DIP SOLDER<br>  <br><br>1 6 (.064 IN ) P.C. BOARD | F | RIGHT ANGLE DIP SOLDER<br>  <br><br>1 6 ( 064 IN ) P C BOARD |
| G | RIGHT ANGLE DIP SOLDER<br>  <br><br>1 6 (.064 IN ) P.C. BOARD | J | RIGHT ANGLE DIP SOLDER<br>  <br><br>1 6 ( 064 IN ) P C BOARD |
| P | RIGHT ANGLE DIP SOLDER<br>  <br><br>3 2 ( 125 IN ) P C BOARD | Q | RIGHT ANGLE DIP SOLDER<br>  <br><br>3 2 ( 125 IN ) P C BOARD |

FIGURE 2. Connector termination styles.



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# TERMINAL STYLES

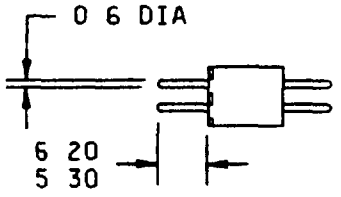
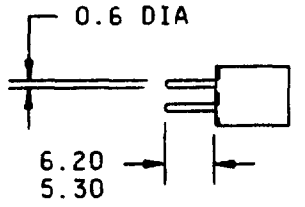
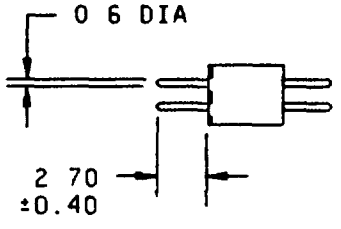
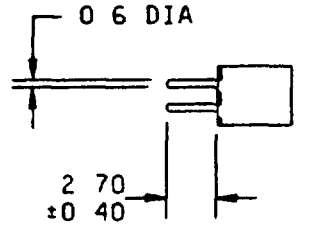
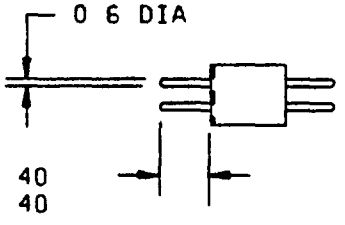
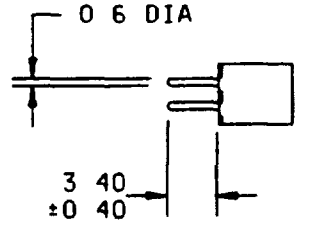
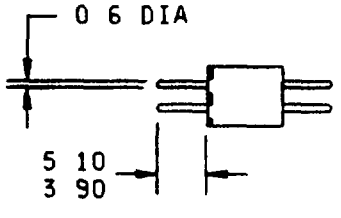
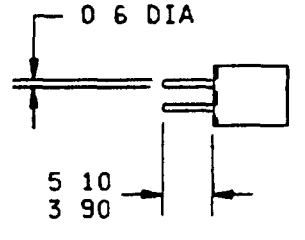
| REF | MALE CONTACTS | REF | FEMALE CONTACTS |
|---|---|---|---|
| A | STRAIGHT DIP SOLDER<br>  | D | STRAIGHT DIP SOLDER<br>  |
| T | STRAIGHT DIP SOLDER<br>  <br>1 6 ( 064 IN ) P C. BOARD | U | STRAIGHT DIP SOLDER<br>  <br>1 6 ( 064 IN ) P C BOARD |
| N | STRAIGHT DIP SOLDER<br>  <br>2 4 ( 094 IN ) P C BOARD | Z | STRAIGHT DIP SOLDER<br>  <br>2 4 ( 094 IN.) P C BOARD |
| V | STRAIGHT DIP SOLDER<br>  <br>3 2 ( 125 IN ) P C BOARD | W | STRAIGHT DIP SOLDER<br>  <br>3 2 ( 125 IN ) P C BOARD |

FIGURE 2 Connector termination styles - Continued.



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# TERMINAL STYLES

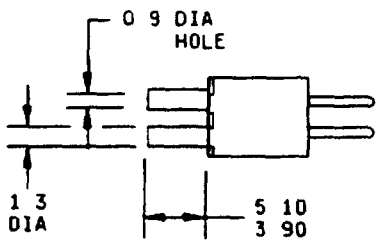
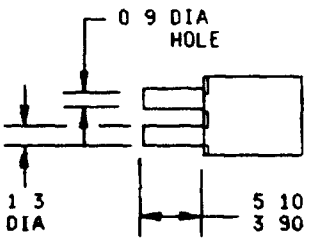
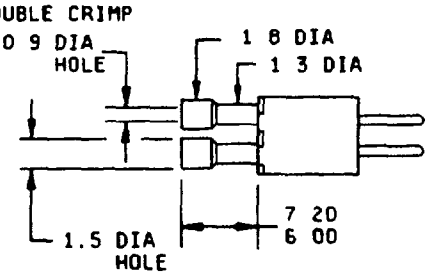
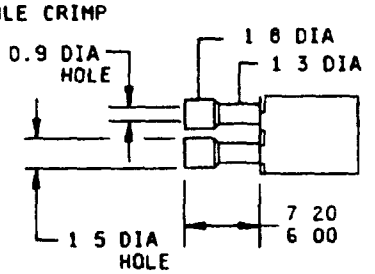
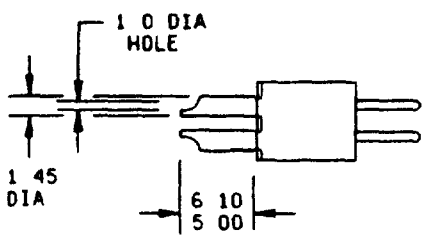
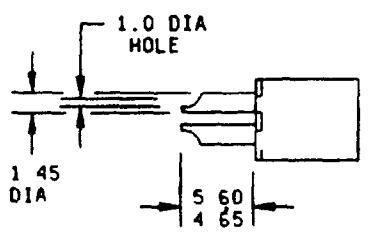
| REF | MALE CONTACTS | REF | FEMALE CONTACTS |
|---|---|---|---|
| R | CRIMP<br>  <br><br>FOR 22,24 AND 26 AWG. | K | CRIMP<br>  <br><br>FOR 22,24 AND 26 AWG |
| M | DOUBLE CRIMP<br>  <br><br>FOR 22,24 AND 26 AWG | H | DOUBLE CRIMP<br>  <br><br>FOR 22,24 AND 26 AWG |
| S | SOLDER CUP<br>  <br><br>FOR UP TO 22 AWG | L | SOLDER CUP<br>  <br><br>FOR UP TO 22 AWG |

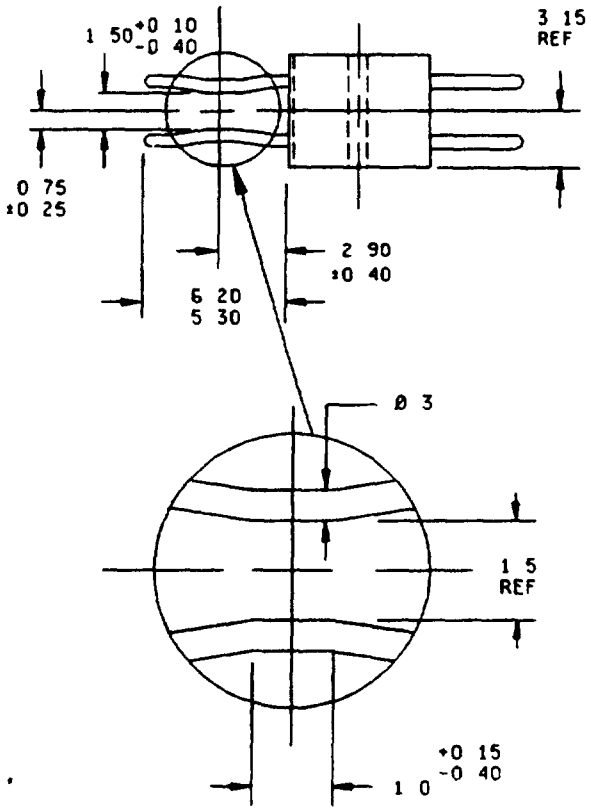
FIGURE 2. Connector termination styles - Continued

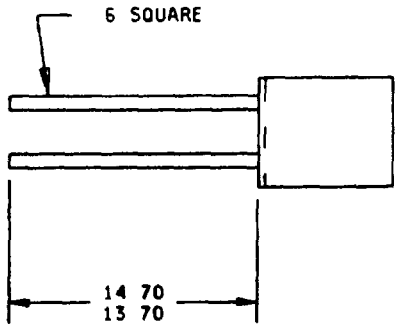


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# TERMINAL STYLES

| REF | MALE CONTACTS | REF | FEMALE CONTACTS |
|---|---|---|---|
| X | SURFACE MOUNT | Y | WRAP POST |



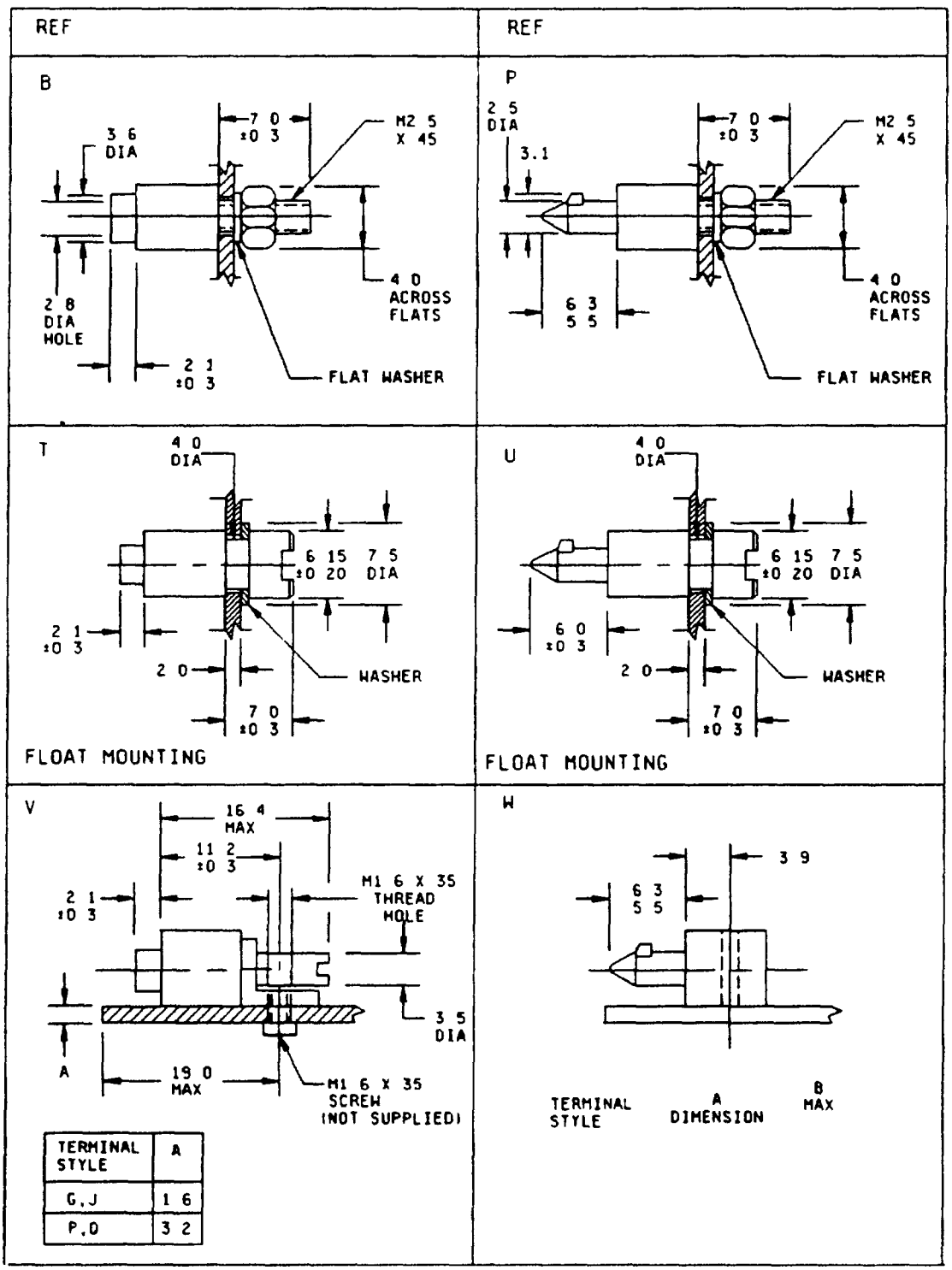

NOTE: Dimensions are in millimeters.

FIGURE 2 Connector termination styles - Continued



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# RECEPTACLE MOUNTING STYLES




| TERMINAL STYLE | A |
|---|---|
| G,J | 1 6 |
| P,D | 3 2 |

NOTE Dimensions are in millimeters.

FIGURE 3 Mounting styles.



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# LOCKING HARDWARE

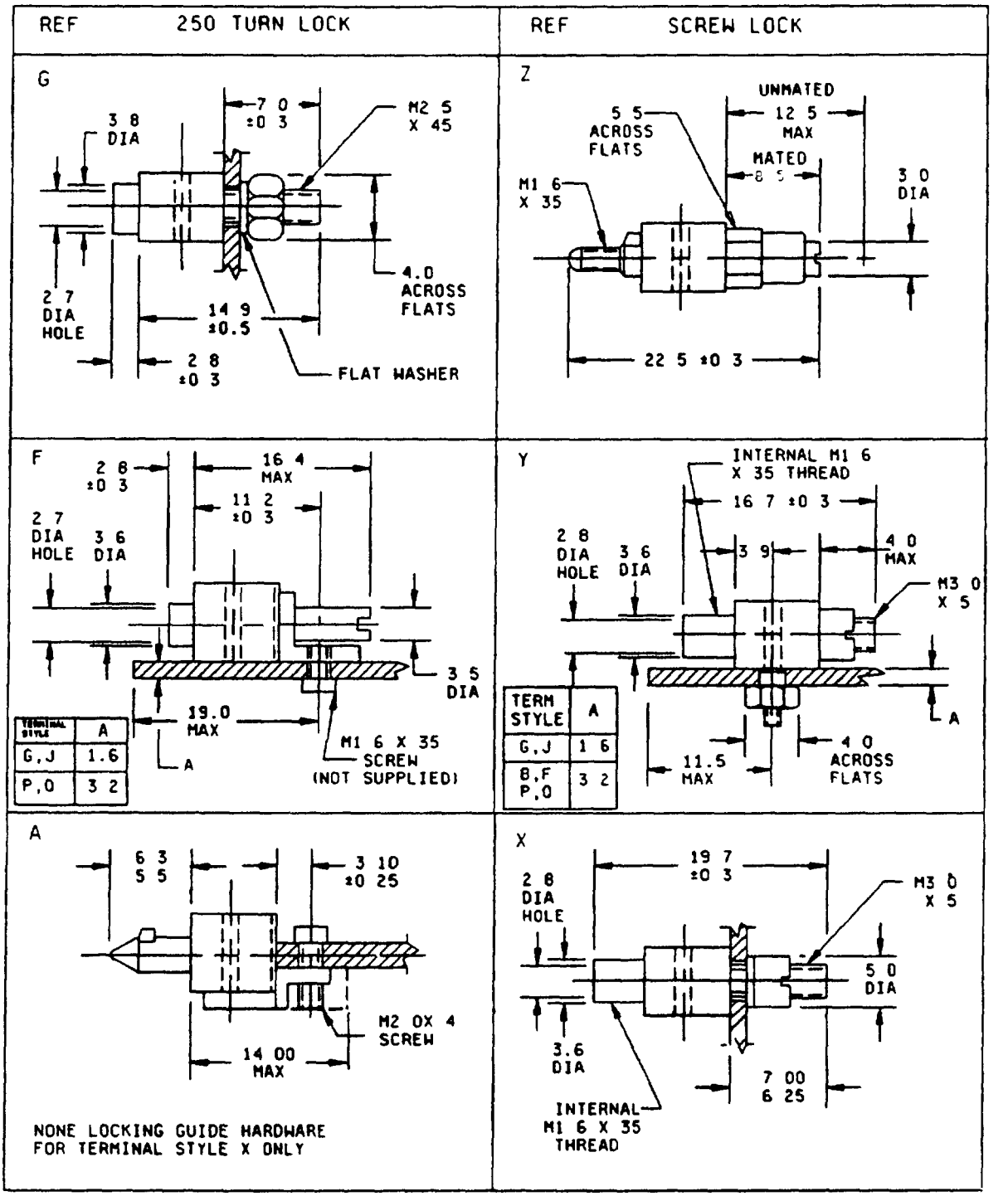
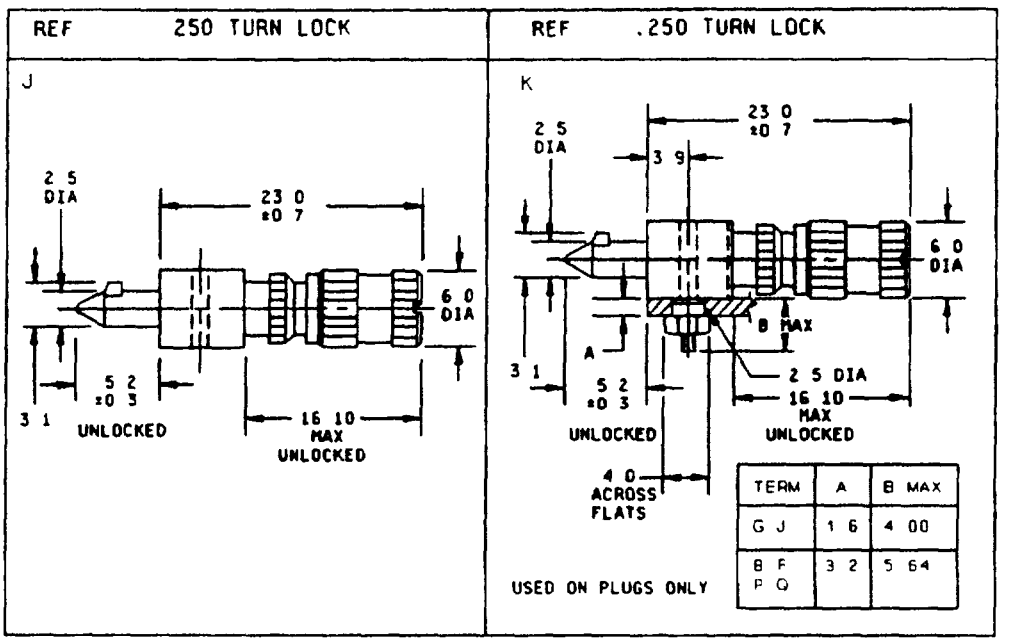



FIGURE 4 Locking hardware



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## LOCKING HARDWARE




| TERM | A | B MAX |
|---|---|---|
| G J | 1 6 | 4 00 |
| B F P Q | 3 2 | 5 64 |

NOTES:

1. Dimensions are in millimeters.
2. This is a metric design
3. Unless otherwise specified, tolerance is ±0.10 millimeter
4. These connectors mate with MIL-C-55302/160.
5. See table II for mating.
6. Contact identification may be located adjacent to cavity on face or side and may be rotated 180°.
7. Crimp contacts are shipped not inserted into insulator.
8. Use crimp tool M22520/2-01 for R and K terminal styles with Daniels manufacturing positioner K547.
9. Use crimp tool M22520/2-01 for M and H terminal styles with Daniels manufacturing positioner K547 and K640.

FIGURE 4. Locking hardware - Continued.

TABLE II. Mating references.

| Type | Will only mate with (see note 4) | Locking method |
|---|---|---|
| B | P, U, W | None |
| F | J, K | Push 1\4 turn |
| G | J, K | Push 1\4 turn |
| J | F, G | Push 1\4 turn |
| P | B, T, V | None |
| T | P, U, W | None |
| U | B, T, V | None |
| V | P, U, W | None |
| W | B, T, V | None |
| X | Z | Screw |
| Z | X, Y | Screw |



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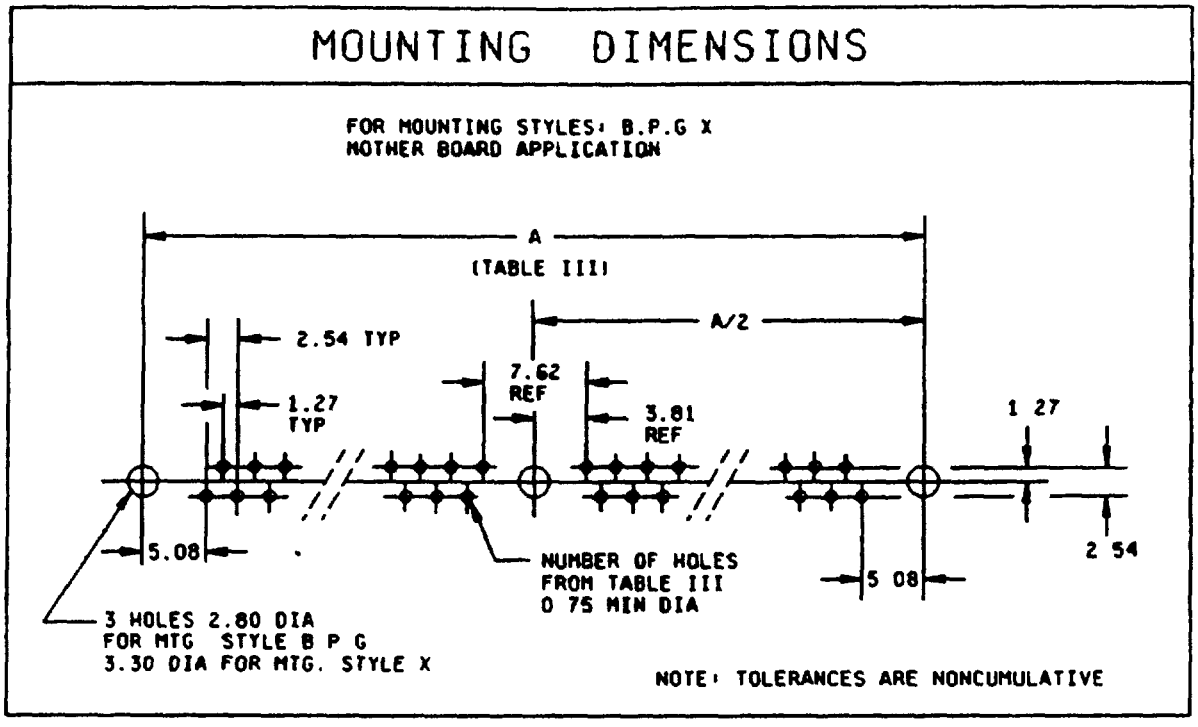


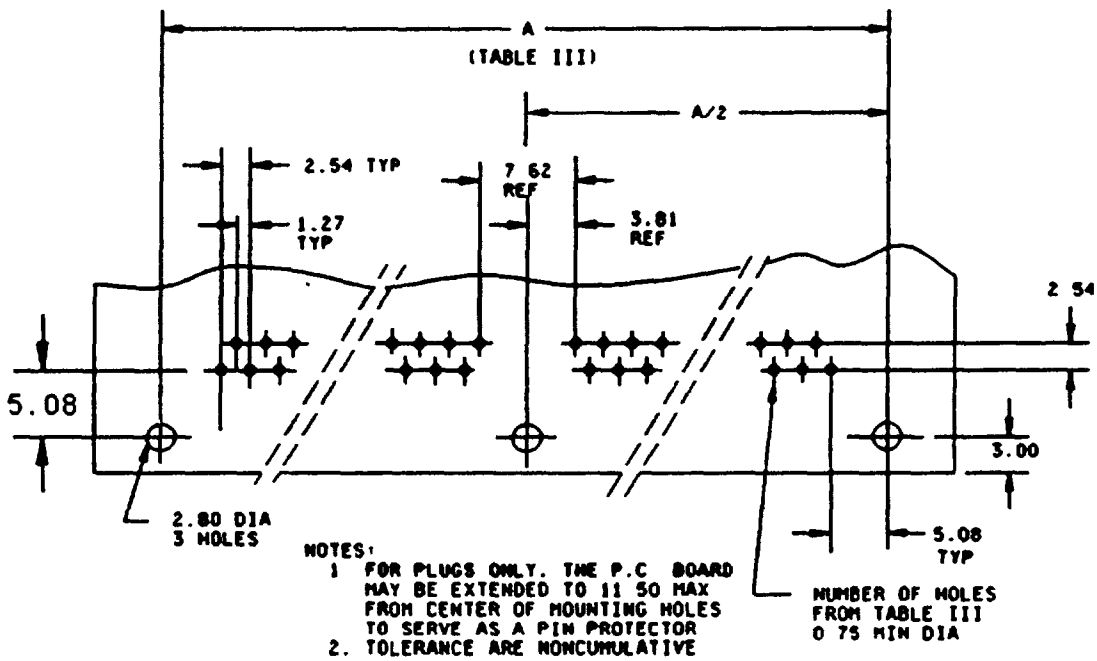

FIGURE 5. Mounting dimensions.



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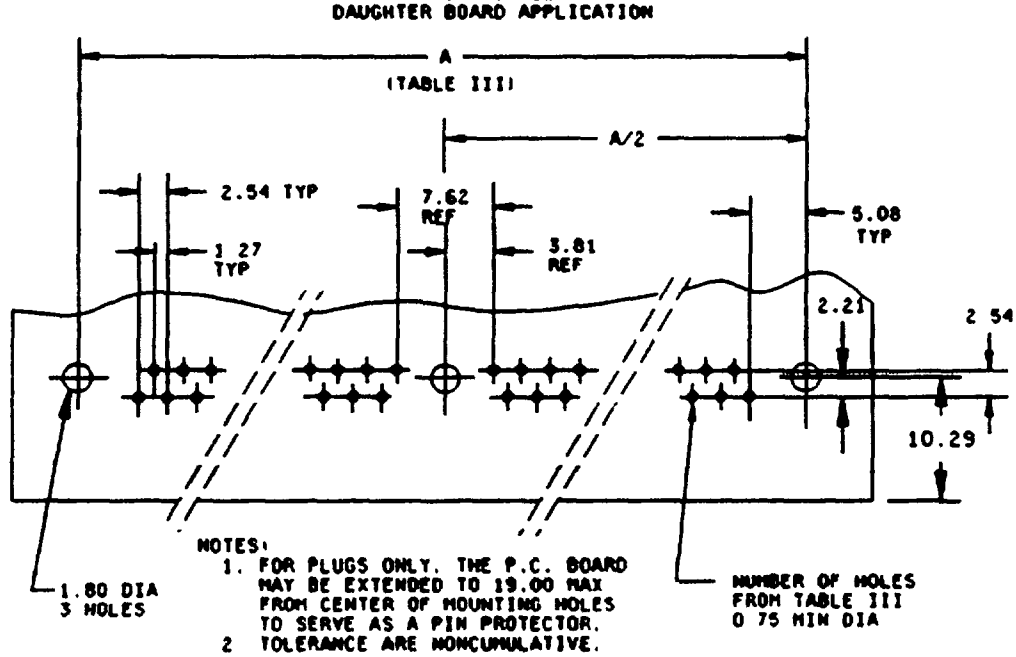
# MOUNTING DIMENSIONS

FOR MOUNTING STYLES: K,M,Y
DAUGHTER BOARD APPLICATION




NOTES:
1. FOR PLUGS ONLY. THE P.C. BOARD MAY BE EXTENDED TO 11.50 MAX FROM CENTER OF MOUNTING HOLES TO SERVE AS A PIN PROTECTOR.
2. TOLERANCE ARE NONCUMULATIVE.

FOR MOUNTING STYLES: F,V
DAUGHTER BOARD APPLICATION




NOTES:
1. FOR PLUGS ONLY. THE P.C. BOARD MAY BE EXTENDED TO 19.00 MAX FROM CENTER OF MOUNTING HOLES TO SERVE AS A PIN PROTECTOR.
2. TOLERANCE ARE NONCUMULATIVE.

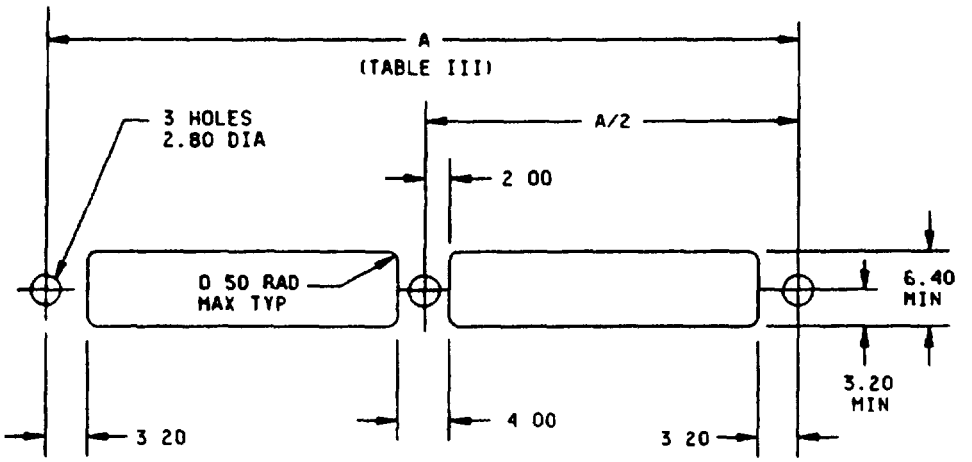
FIGURE 5. Mounting dimensions - Continued



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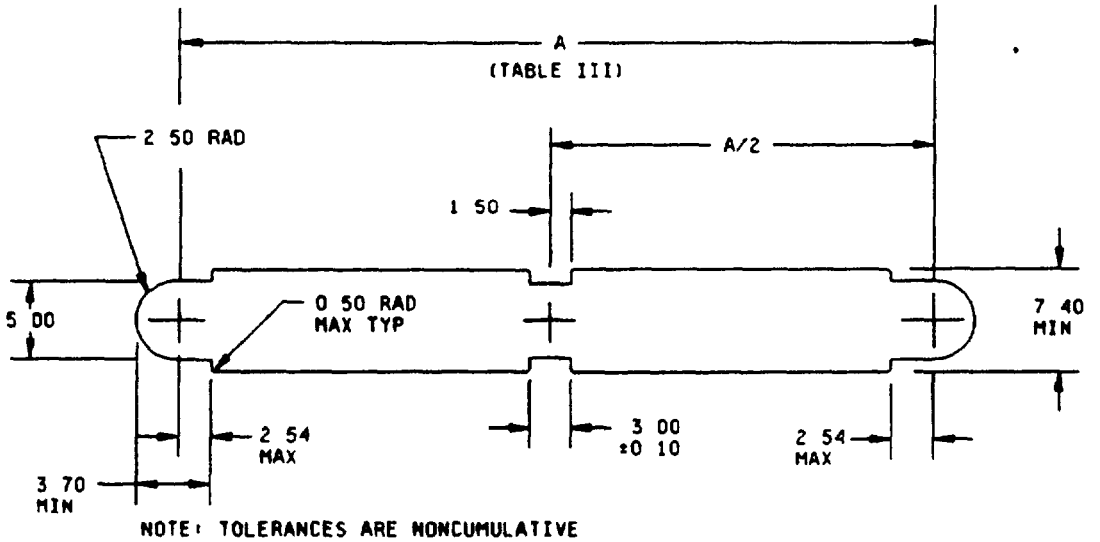
MOUNTING DIMENSIONS

PANEL CUTOUT FOR FIXED
MOUNTING STYLES B,G,P




NOTE: TOLERANCES ARE NONCUMULATIVE

PANEL CUTOUT FOR FLOAT
MOUNTING STYLES T,U

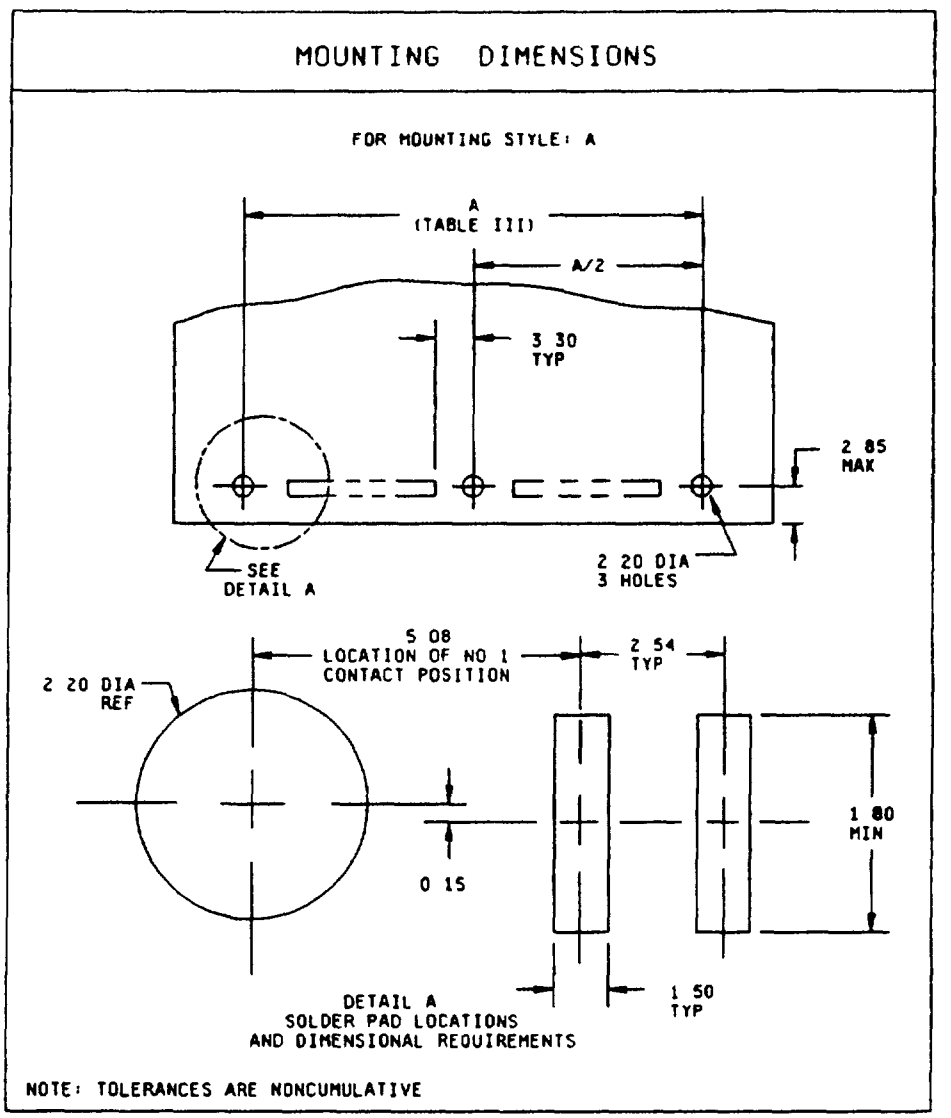



NOTE: TOLERANCES ARE NONCUMULATIVE

FIGURE 5 Mounting dimensions - Continued



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NOTE Dimensions are in millimeters.

FIGURE 5. Mounting dimensions - Continued.

TABLE III. Mounting references.

| Number of contacts | A |
|---|---|
| 72 | 106.70 |
| 84 | 121.90 |
| 96 | 137.20 |
| 120 | 167.60 |



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REQUIREMENTS.

Design and construction

Dimensions and configuration: See figures 1 through 5

Material:

Insulators: Diallyl phthalate in accordance with MIL-M-14.

Guides and hardware: Brass in accordance with ASTM B36, B16, B124, B453, or B455; nickel plated in accordance with QQ-N-290 or AISI 300 series stainless steel or equivalent in accordance with QQ-S-763 or ASTM A582, passivated in accordance with QQ-P-35

Plating: Plating in accordance with MIL-C-55302 or as noted below.

Mating surface (pin and wire): Gold in accordance with MIL-G-45204, class 1, type II, grade C over nickel in accordance with QQ-N-290, 30-150 microinches.

Terminations: Solderable areas (including crimp barrels and wire wrap terminations) shall be gold in accordance with MIL-G-45204, type III, grade A, 10 microinches minimum thickness or gold in accordance with MIL-G-45204 type II, grade C, class 00 over nickel in accordance with QQ-N-290, 30 to 150 microinches thick. Solderable areas must meet the requirements of MIL-STD-202, method 208.

All other surfaces: Gold in accordance with MIL-G-45204, type II, grade C, 5 microinches minimum over nickel in accordance with QQ-N-290, 30-150 microinches

Contact identification: See figure 1 and table I.

Contact separation forces:

Minimum force: 0.14 newton (0.50 ounce) with a 0.5800 millimeter (0 0228 INCH)+0.005/-0.000 diameter pin after being cycled 3 times.

Note: Steel pin test surface roughness to be 0.025 - 0.25 micron rms.

Unmating: The maximum force of a connector assembly shall not exceed 0 53 newton (1.90 ounces) multiplied by the number of contact positions.

Mating The maximum force of a connector assembly shall not exceed 0.70 newton (2.5 ounces) multiplied by the number of contact positions.

Contact retention minimum: Functional value of 22.2 newtons (5 pounds); destructive test 35.3 newtons (8 pounds) maximum design and qualification value.

Contact rating. 5 amperes per line

Durability· 2,000 cycles minimum.

Dielectric withstanding voltage:

Sea level: 1,400 V ac rms test.

High altitude· 300 V ac rms test.

Keying: Rotation of guides; 6 positions each, 36 possible combinations.



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Connector Part or Identifying Number (PIN)

Example. D55302/ 163 B 120 W

- Specification number
- Specification sheet number
- Type of contact and termination (see figures 1 and 2)
- Number of contacts (see table I)
- Guide hardware number (see figures 3, 4, 5 and tables II and III)

Revision letters are not used due to the extensiveness of the changes.

## CONCLUDING MATERIAL

Custodian.
Army - CR

Review activities:
Army - AR, AT, AV, ME, MI
DLA - ES

Preparing activity:
Army - CR

Agent:
DLA - ES

Project 5935-A418-05)



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